

Indicators of living standards

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There are a variety of indicators that can be used to assess the living standards in a country. These include the number of people or households that own a given consumer good, such as cars, laptops, mobile (cell) phones and TVs (see Table 32.1). The number of patients per doctor, enrolment in tertiary education, the adult literacy rate, the average food intake per person and conduct of free elections in the country can also be examined. A main measure of living standards, however, is still probably real GDP per head. There are also composite indicators such as the Human Development Index (HDI) and the Genuine Progress Indicator (GPI).

Cars per 1000 population		Internet users per 100 population		Mobile phone subscriptions per 100 people	
Puerto Rico	805	Iceland	98.2	Macau	322.6
Luxembourg	746	Bermuda	96.8	Hong Kong	233.6
Iceland	720	Norway	96.3	Kuwait	218.4
Malta	665	Denmark	96.0	Saudi Arabia	179.6
Bahrain	662	Andorra	95.9	UAE	178.1
New Zealand	648	Liechtenstein	95.2	Bahrain	173.3
Lithuania	623	Luxembourg	94.7	Kazakhstan	172.2
Canada	612	Netherlands	93.2	Gabon	171.4
Italy	607	Sweden	92.5	Suriname	170.6
Finland	592	Finland	92.4	Botswana	167.3
		Monaco	92.4		

Table 32.1: The top ten countries in terms of highest car ownership, internet users and mobile phone subscriptions 2014

Source: Pocket World in Figures, 2017 edition, pages 76, 98 and 99, The Economist

Real GDP per head as an indicator of living standards

An increase in real GDP per head would suggest that living standards have risen, but it may not necessarily indicate the real situation for a number of reasons. One is that real GDP is an average. Not everyone may benefit from a rise in the average income level. The extra income may be unevenly spread with a few receiving much higher income and some not receiving any extra income.

Higher output will obviously mean that more goods and services are being produced, but not all of these may add to people's living standards. For example, an increase in the output and consumption of tobacco may actually reduce the quality of people's lives by affecting

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their health and life expectancy adversely. A rise in police services, due to a higher crime rate, is also unlikely to improve most people's living standards.

Increases in real GDP per head figures may understate the products available to people due to undeclared economic activity and non-marketed output. It may, however, overstate it if the quality of output is falling. Living standards are also influenced by other factors besides the material goods and services produced. If output rises but working conditions deteriorate, the number of working hours increases, or pollution increases, people may not feel better off.

Comparing living standards between countries

Again, one of the main measures is real GDP per head. This measure has the advantage that it takes into account differences in population size and also incorporates adjustments for inflation.

There are still a number of reasons why it cannot serve as a definite way of ascertaining the quality of living standards of people in a country. (In fact, some of the citizens of some countries enjoy a better quality of life than some of those in richer countries.) This is because there can be differences in the distribution of income, the size of the informal economy, working hours and conditions, the composition and quality of output and environmental conditions between countries.

There is also a potential problem in comparing different countries' real GDP per head because countries measure their output in terms of their own currency initially. The comparison, on the other hand, requires to be done in a common unit. There is a risk that if an unadjusted currency is used, the comparison may be distorted. This is because the value of a currency can change on an hour to hour basis. For example, US dollars may be used. If the real GDP of Kenya was KSh1200bn and the exchange rate was initially \$1 = 100 Kenyan Shillings (KSh), the GDP of Kenya would be valued at \$12bn. If the exchange rate changed to \$1 = KSh80 the next day, the value of its output in dollars would change to \$15bn, despite the fact that in that period Kenya would not have increased its output by 25%.

It is for this reason that economists make use of purchasing power parity (PPP) exchange rates when comparing countries' GDP. These use an exchange rate based on the buying power of currencies in their own countries. They compare the amount of a country's currency needed to buy the same basket of goods and services with that of another country or countries' currencies. If in the USA a given basket of products sells for \$5000 and in Kenya, it sells for KSh45 000, the exchange rate used would be \$1 = KSh9. This figure would not be affected by market changes in the price of currencies on the foreign exchange market.

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The Human Development Index

The Human Development Index (HDI) was developed by a team of economists led by Dr Mahbub ul Haq, while he was working for the United Nations Development Programme (UNDP). It has been published every year since 1990.

The HDI is a wider measure than real GDP per head. Besides including GDP per head (see Note, right), the HDI considers two other indicators of living standards. One is the length of time for which people can enjoy life, measured as life expectancy at birth. The second is education. This is measured by mean years of schooling and expected years of schooling. On the basis of their HDI values, countries are categorised into very high human development, high human development, medium human development and low human development. The HDI shows that economic growth and human progress may not always match. Some countries such as Namibia and South Africa have recently enjoyed a higher ranking in terms of GDP per head than HDI, whilst others including Costa Rica and Cuba usually score more highly in terms of HDI than GDP per head.

Although the HDI does take into account other factors that influence people's living standards, it has been criticized for what it leaves out. In fact, it has been stated that the index would be high for someone living for a long time in a prison who has been well educated. Among the factors it does not take into account are political freedom and the environment. It also does not consider differences in life expectancy, education and differences in income between males and females and between those living in rural and urban areas, among other groups. Table 32.2 shows the countries with the highest and lowest HDI rankings in 2014.

Ranking	Top ten countries	Ranking	Bottom ten countries
1	Norway	178	Mali
2	Australia	179	Mozambique
3	Switzerland	180	Sierra Leone
4	Denmark	181	Guinea
5	Netherlands	182	Burkina Faso
6	Germany	183	Burundi
6	Ireland	184	Chad
8	USA	185	Eritrea
9	Canada	186	Central African Republic
10	New Zealand	187	Niger

Table 32.2: HDI rankings in 2014

Table 1 page 208 Human Development Report 2015 UNDP

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NOTE: The HDI now actually uses Gross National Income per head rather than GDP per head. This is very similar to GDP per head although it includes workers' remittances. For consistency, GDP per head is used here.

The Genuine Progress Indicator (GPI)

The Genuine Progress Indicator (GPI) is another composite measure containing a number of indicators. It was developed in 1995. It starts with GDP. It then adjusts for income distribution. As the poor receive more benefit from a rise in income than the rich, the GPI rises when the poor receive a higher proportion of income and it falls when income becomes more unevenly distributed. Then, it makes a number of deductions and additions. The items deducted are those which reduce living standards now or in the future. Examples of such items include the costs of crime, traffic accidents, carbon dioxide emissions, depletion of non-renewable resources and the loss of wetlands and forests. Items which are thought to make a positive contribution to current or future living standards are added. These include the value of housework and volunteer work and increases in leisure time.

Other measures

There is a wide range of other measures of living standards. Whilst the HDI is the UNDP's best known measure, it also produces several other measures. Another of the UNDP's measures is the Gender Inequality Index (GII) which considers gender-based disadvantage in terms of reproductive health, empowerment and the labour market. The health dimension is measured in terms of maternal mortality rate and adolescent fertility rate. There are two measures to the empowerment dimension which are the percentage of seats in the national parliament held by women and the relative percentage of men and women with at least a secondary education. Involvement in the labour market is assessed by comparing the labour force participation rates of men and women.

Economists also calculate the Happy Life Expectancy Index (HLEI). This seeks to measure the degree to which people live long and happy lives. It is found by multiplying life expectancy at birth by a happiness index. Bhutan also measures Gross National Happiness. This is based on a survey which assesses people's happiness using thirty-three indicators in nine categories. One category is named living standard and this has the three indicators of income per head, assets and housing. The other eight categories are psychological wellbeing, health, time use, education, cultural diversity and resilience, good governance and community vitality, ecological diversity, and resilience.

Comparing living standards and income distribution

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Income is unevenly distributed between households in every country, but to different extents. Wealth is even more unevenly distributed. Wealth is a stock of assets which have a financial value. Some of these assets, such as shares and government bonds, give rise to income. Some people have a considerable amount of wealth, whilst others have none.

Measures of income and wealth inequality

To assess the distribution of income and wealth in a country, economists calculate the percentages of income earned and wealth owned by given proportions of the population. Two common proportions used are deciles (tenths) and quintiles (fifths).

The causes of differences in living standards and income and wealth inequality

Living standards vary between countries and between regions within a country for a large number of reasons. These include, for example, differences in income, wealth, education and healthcare systems, levels of pollution and working hours.

Income may be unevenly distributed between households within and between countries due to uneven holdings of wealth, differences in the composition of households and differences in the opportunity and ability to earn an income. Some wealthy households can live off the income their wealth generates. In more cases, the income from wealth adds to earned income.

Households with a number of workers are likely to have a higher income than those with one or no workers and a high number of dependants.

The wages received by workers are influenced by their skills, qualifications and the number of hours for which they work. High skilled workers with better qualifications are likely to be in high demand and hence are likely to receive high wages. Full-time workers usually earn more than part-time workers. Some people may be dependent on state benefits or help from relatives and their income is likely to be relatively low.

Wealth is unevenly distributed because there are differences in the assets inherited by people, their savings and entrepreneurial skills. In fact, inheritance is a major reason for some people being wealthy. The more a person can save, the wealthier they will become. Of course, in this sense wealth creates wealth. Wealthy people can afford to save more and this makes them wealthier. People with an entrepreneurial flair may be able to build up a business from scratch and become wealthy.

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Absolute and relative poverty

Absolute poverty occurs when people do not have enough income to pay for their basic needs. They may lack access to, for example, adequate food, clothing, shelter, sanitation and healthcare. People experience relative poverty, when they are poor relative to other people in the country. People who are relatively poor are unable to participate fully in the normal activities of the society they live in. People who are regarded as relatively poor in a rich country may not be seen as such if they had the same income in a poor country.

The causes of poverty

There are a number of reasons why people are poor. These include being unemployed, being in low-paid work, falling ill and growing old. In all of these cases, people may lack the income to ensure that they do not experience poverty. Once in poverty, it can become difficult to get out of it. People can become trapped in a vicious circle of poverty. Those who are poor are likely to have worse than average education and healthcare. This will reduce their productivity, employment opportunities and income, and will also affect the prospects of their children.

The Multidimensional Poverty Index (MPI)

One measure of poverty is the Multidimensional Poverty Index (MPI). This index is based on deprivations in education, health and standard of living. Household surveys are undertaken to find, for example, how many households have no member who has completed five years of schooling, someone who is malnourished, and who has no access to clean water and no electricity.

There are two indicators in the health category - child mortality and nutrition, each with a weighting of $1/6$ th. There are again two indicators in the education category with a weighting of $1/6$ th each. These are years of schooling and child school attendance. The total weighting given to standard of living is one-third, as it is with education and health. This time, however, the category is subdivided into six rather than two indicators. These are electricity, improved sanitation, safe drinking water, flooring, cooking fuel and assets.

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Each part has a weighting of 1/18th. A household is regarded as poor if it suffers deprivations on a third, or more, of the ten indicators (see Figure 33.1).

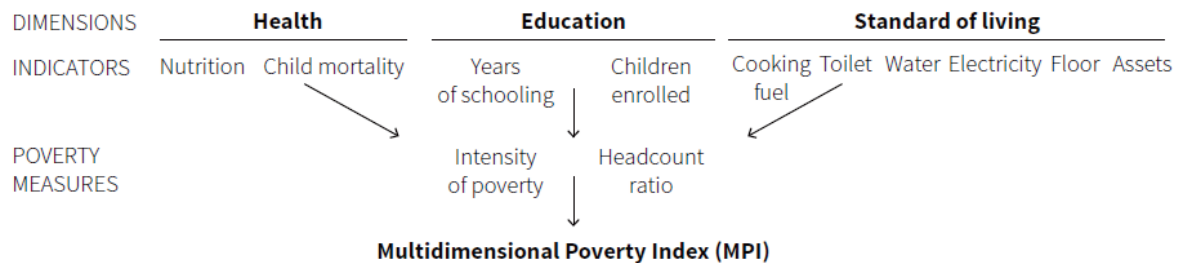


Fig. 33.1: The Multidimensional Poverty Index

Source: United Nations Development Programme, Human Development Reports, Multidimensional Poverty Index (MPI), 2016, hdr.undp.org/en/content/multidimensional-poverty-index-mpi

Possible government policy measures to reduce poverty

There are a number of measures a government may take in a bid to reduce poverty. Its choice of measures, however, can be restricted by the tax revenue it has raised to fund them. Among the possible measures are:

- Improving the quantity and quality of education. In the long term this can be a very effective policy, as it can increase the job prospects and earning potential of the poor and their children.
- Promoting economic growth. For example, increasing government expenditure or reducing the rate of interest, will increase aggregate demand. This can increase output and create jobs. Unemployment is a major cause of poverty. Having unemployed and underemployed workers not only lowers their living standards, but also the living standards of others since output will be below potential.
- Introducing or raising a national minimum wage. This is designed to tackle the problem of people experiencing low living standards due to low wages.
- Encouraging more multinational companies to set up in the country. The opening up of new firms in the country should create more employment opportunities.
- Providing benefits or more generous state benefits. The elderly, and some of the sick and disabled, may not be able to work and may not have any savings to support them. Giving them benefits, or raising the benefits they receive, may enable them to avoid absolute poverty. What is more debatable is the effect that raising unemployment benefit will have on poverty. If there is a lack of jobs, it may help in the short run, as it will not only raise the living standards of the unemployed, but may also reduce unemployment by increasing aggregate demand. If, however, jobs are available, and the unemployed are not filling them because they receive a higher income on benefits, raising benefits will reduce the incentive to work.

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- Land reform. In a number of countries, many agricultural workers work on land owned by a few rich land owners. Those who rent or lease land may be reluctant to improve the fertility of the land. In such circumstances, making ownership of land more equal may increase output and living standards.

Measures to raise living standards

Measures to reduce poverty should also raise general living standards. Improving education and training will enhance the knowledge and earning potential of the people, and their ability to participate in the political system of the country. Reducing unemployment, as already mentioned, also raises living standards by increasing the quantity of available goods and services.

Other ways of improving living standards include improving healthcare, increasing and improving the housing stock, improving the working conditions and reducing pollution. Economists debate about the extent to which government intervention is needed to achieve these objectives. Some economists argue that government policies will be needed, such as legislation, to give workers holiday entitlement and government provision of housing. Other economists argue that the private sector is better at providing improved living standards. If this is the case, living standards might be increased by reducing government regulation and taxation. For example, if corporation tax is reduced, firms may be encouraged to expand taking on more workers and providing more training.

Government policies on the distribution of income and wealth

Governments may decide to influence the distribution of income and wealth because of concerns that a very uneven distribution may be socially divisive. It may also want to ensure that everyone has access to a certain standard of living. In influencing the distribution, however, governments may be concerned that it does not reduce incentives to entrepreneurs and workers.

Governments can influence the distribution in a number of ways including:

- taxation
- the provision of cash benefits
- the provision of free state education and healthcare, and
- using labour and macroeconomic policies.

Progressive taxes make the distribution of income and wealth more even. The provision of unemployment and other cash benefits can help maintain a reasonable standard of living. Provision of free state education and healthcare can ensure that everyone has access to these essential services, and it may also offer the people an opportunity to improve their

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living standards. Other government policies that can affect distribution of income include minimum wage legislation, regional policy and measures to reduce unemployment.

Factors that affect population growth

The size of a country's population can grow as a result of a natural increase or net immigration. A natural increase occurs when the birth rate exceeds the death rate. For example, the birth rate in Iran in 2016 was 15.6 per 1000 and the death rate was 4.6 per 1000, giving a natural rate of increase of 11 per 1000 or 1.1%. The birth rate is influenced by the average age of the population, the number of women in the population and women's fertility rate (that is, the average number of children per woman).

The death rate is influenced by nutrition, housing conditions, medical care, and lifestyles, working conditions, involvement or non-involvement in military action.

Net immigration occurs when more people come into the country to live (immigrants) than people who leave it to live elsewhere (emigrants).

The reasons for different rates of population growth

The birth rate is likely to be high when there is a young average aged population in which women marry young, infant mortality rate is high, women are not well-educated, most women do not work and it is cheap to bring up children. Other factors leading to a high birth rate are the lack or disapproval of family planning, government cash incentives to have children, but a lack of government help to care for the sick and elderly.

In contrast, in a country where it is expensive to have children (because there is a legal requirement to send children to school for a number of years and an expectation that some of them will go on to higher education), and well-paid jobs are open to women, the birth rate is likely to be low. If the government does provide state pensions, sickness and disability benefits, families may not feel that they have to have a large family in order to support them.

A country in which people have healthy diets, enjoy good housing facilities and access to high quality medical care, do not smoke or consume excess alcohol, exercise regularly, enjoy good working conditions and the country is at peace with other countries, is likely to have a relatively low death rate.

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The rate and pattern of net migration is influenced by relative living standards at home and abroad, persecution of particular groups and extent of control on the movement of people. If living standards abroad are better (or perceived as better), there is persecution at home and no restriction to immigration by other countries, some people are likely to emigrate. Most migrants tend to be single people, of working age.

In deciding what is happening to a country's population size, the relationship between the birth rate, death rate and net migration have to be considered. A fall in the death rate, for example, will not necessarily lead to an increase in population if it is offset by a fall in the birth rate and/or net emigration. Latvia's population, for example, is declining because its birth rate has fallen at a greater rate than its death rate and it is currently experiencing net emigration.

The effects of changes in the size and structure of populations

Population structures

The gender distribution of a population indicates the number of males, compared to the number of females. In most countries, more boys are born than girls. However, in a number of countries females outnumber males. This is largely due to women, on an average, living longer than males and higher male infant mortality rates. In some countries, such as the United Arab Emirates (UAE) and China, there are more males in the population than females. This is because some families in China may favour males, while in the UAE more males than females from other countries come to work and live in the country.

The age distribution is the division of the population into different age groups. In broad terms, the categories are people under 16, those between 16-64 and those over 65.

Diagrams in the form of bar charts, referred to as population pyramids, show a more detailed breakdown of the different age groups. The traditional view of the population pyramid of a country with relatively low development is a pyramid shape such as that shown in Figure 34.1.

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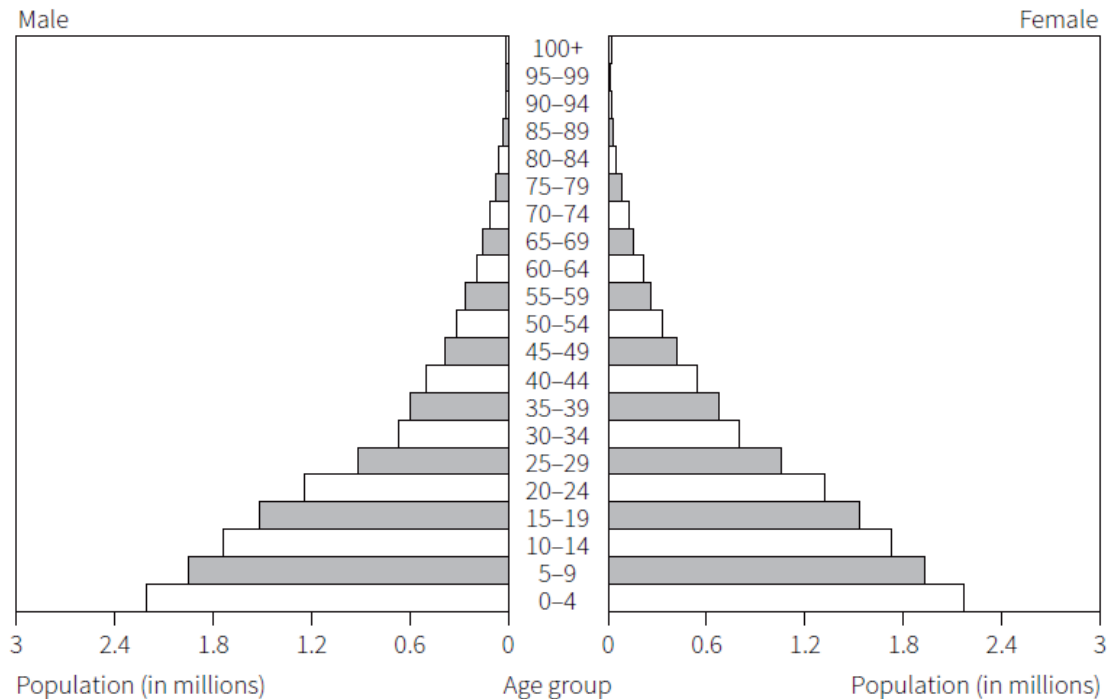


Fig. 34.1: Population pyramid, Mozambique, 2016

Source: CIA World Factbook

Such a pyramid reflects a high birth rate and a high death rate. It has a large proportion of young people in its population with only a small percentage of people reaching older age groups. Of course, not all countries with relatively low development fit this pattern of population structure.

The traditional view of the population pyramid of a country with high development is reflected in Figure 34.2. This is based on a lower birth rate and death rate with more people surviving till they enter old age.

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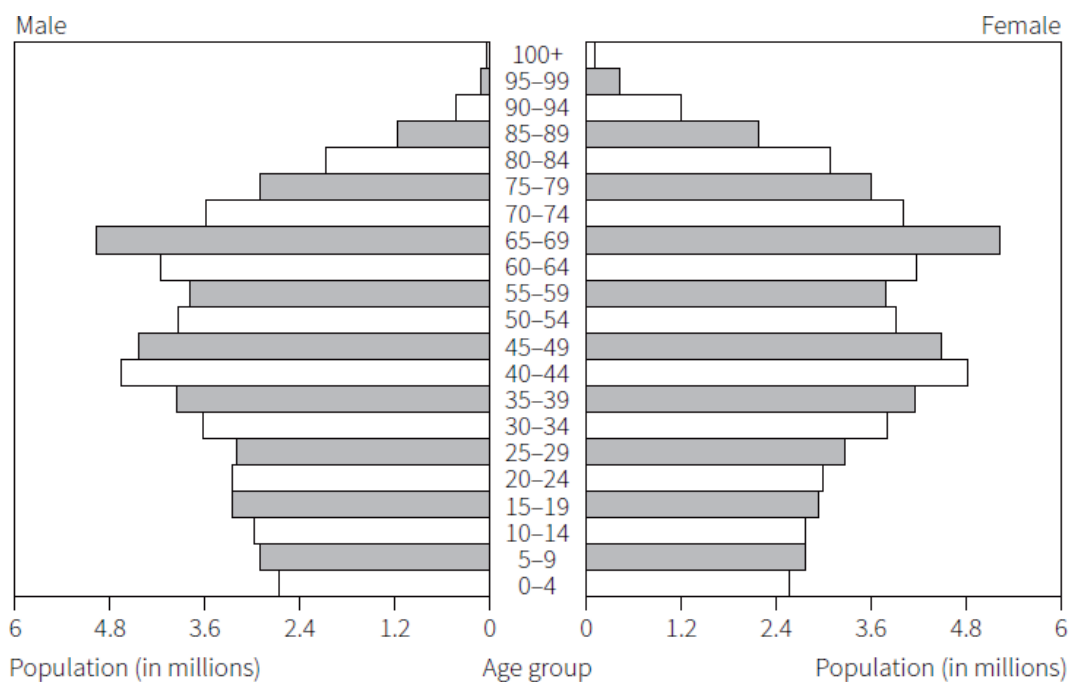


Fig. 34.2: Population pyramid, Japan, 2016

Source: CIA World Factbook

The age structure of a population influences its dependency ratio. This is the:

$$\frac{\text{Number in dependent age groups}}{\text{Number in the labour force}} \times 100$$

The dependent age groups are those below school leaving age and those above retirement age.

The optimum population

The term optimum population refers to the number of people which, when combined with the other resources of land, capital and existing technical knowledge, gives the maximum output of goods and services per head of the population.

The concept is not based just on the number of people in a country or per square kilometer. The balance between population and resources is of crucial importance. A country with a large geographical area and a small population may still be considered as overpopulated if there is a shortage of land, capital and technical knowledge, relative to the number of workers. In such a case, a government may seek to move towards its optimum population either by introducing measures to reduce its population size or by seeking to increase investment. A country is said to be under-populated if it does not have enough of human resources to make the best use of its resources. In such a case, its

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government may encourage immigration. Figure 34.3 illustrates the concept of optimum population.

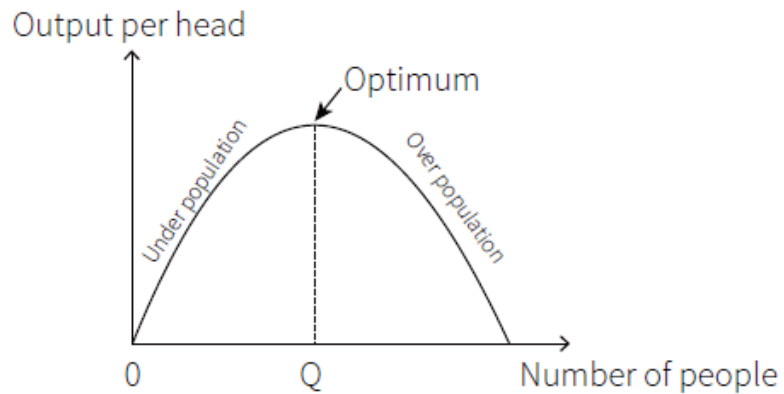


Fig. 34.3: The concept of optimum population

In practice, it is difficult to determine a country's optimum population and size of actual population relative to it. This is, in part, because the quantity and quality of resources are changing all the time.

The effects of an increase in population

The consequences of a growth in a country's population depend on its cause, size of population relative to optimum population and the rate of population growth. Possible benefits of an increasing population are listed below:

- If the population is below the optimum size, the country will be able to make better use of its resources.
- The size of markets will increase. This should enable firms to take greater advantage of economies of scale.
- There may be an increase in factor mobility if the rise has resulted from an increase in the birth rate or immigration. Expanding industries can recruit new workers to the labour force. These people are likely to be familiar with new ideas and methods. If this is the case, firms' training costs will be reduced.
- Extra demand will be generated. This is likely to stimulate investment and this may lead to the introduction of new technology.
- A rise in the labour force now due to net immigration and in the future, caused by a rise in the birth rate. Net immigration will bring in more workers. More children being born will increase the dependency ratio in the short term but in the long term, will result in more workers.

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Despite the possible advantages of an increasing population, there are a number of reasons for a government to be concerned about population growth. These include:

- Concerns about famine. If a country is currently overpopulated and agricultural productivity is low, there is a risk that the country may not be able to feed more dependants.
- Restrictions on improvements in living standards. Resources which could have been used to improve living standards may have to be devoted to the provision of goods and services for the extra number of dependants in the population. There may be, for example, provision of more healthcare facilities but healthcare services per head may fall.
- Overcrowding. Increases in population may put pressure on housing and social capital and cause traffic congestion.
- Environmental pressure. More people in a country may result in damage to wildlife habitats, water shortages and the depletion of non-renewable resources.
- Pressure on employment opportunities. If there is an increase in the number of people of working age who lack appropriate skills, the government may have to devote more resources to education and training. It must be remembered, however, that immigration in itself does not cause unemployment. The number of jobs in existence is not fixed. Although immigration will increase the supply of labour, it will also result in an increase in aggregate demand.
- Balance of payment pressures. More dependants in the population may result in a rise in imports and some products may need to be diverted from the export to the home market.

Ways of reducing the birth rate

A country concerned that its population is growing too rapidly would not want to raise its death rate! It may however, seek to reduce immigration and try to reduce the country's birth rate. There are a number of ways it could do the latter. An effective way is thought to be improvement of the educational and employment opportunities for women. Educated women tend to be more aware of contraceptive methods, marry later and are likely to restrict their family size, in order to be able to combine raising children with work and also because they have higher hopes for their children's future. Better information and increased availability of family planning services will make it more likely that households will be able to limit their family sizes.

Improvement of healthcare and nutrition reduces infant mortality and, in turn, the birth rate. One reason which may explain a higher number of children in families is the apprehension and concern over their survival. Another reason for having a large family is the need for support in old age. Setting up pension and sickness insurance schemes will

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reduce the need for family support. A government can raise the cost of having children by raising the school leaving age. It can also reduce or stop any financial support it gives to families for each child they have and instead provide incentives for families who restrict the number of children. The most extreme measure is to make it illegal to have children above a specified number.

The consequences of an ageing population

A number of countries, including Japan, the USA and many of those in the EU, are experiencing an ageing population. An ageing population is one in which, as its name suggests, the average age is rising. In Japan, for example, the average age of the population in 2016 was 43 and 26.3% of the population was aged over 60.

An ageing population can be caused by a fall in the birth rate, a fall in the death rate, net emigration or a combination of the three. There are a number of consequences of an ageing population including:

- A rise in the dependency ratio. If people are living longer and there are fewer workers because of net emigration, there will be a greater proportion of consumers to workers.
- A change in the labour force. Older workers may be geographically and occupationally less mobile. They may, however, be more experienced, reliable and patient.
- Higher demand for healthcare. The elderly place the greatest burden on a country's health service.
- Greater need for welfare services, such as caring for the elderly at home and in retirement homes.
- Rise in cost of state and private pensions.
- Change in the pattern of demand. For example, demand for housing for retired people will rise.

Ways of coping with an ageing population

Having a higher proportion of elderly workers will put pressure on governments to spend more. If they do so, it will involve an opportunity cost. The money used, for example, may have been spent on education instead. An ageing population may also raise the tax burden on workers. There are a number of ways a government may seek to reduce these effects. One is to raise the retirement age. This will reduce the cost of pensions and increase tax revenue without increasing tax rates. In a country in which life expectancy is rising, there is some justification for raising the retirement age. Working for a longer period can also increase a worker's lifetime income and, depending on the nature of their work, may keep them physically and mentally healthy for a longer period.

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A government may also try to cope with the financial burden of an ageing population by encouraging or making it compulsory for workers to save for their retirement. It may try to raise the productivity of workers by means of education and training. This may involve an increase in government expenditure but, if successful, may generate more tax revenue. In addition, it may encourage the immigration of younger skilled workers by issuing more work permits. This will reduce the dependency ratio at least in the short term.

Internal migration

Workers tend to migrate from rural to urban areas in search of better jobs and higher incomes. Such migrants may supply growing industries and result in a better allocation of resources. Their movement may also raise living standards in rural areas if initially there had been underemployment and if they send back money to their relatives. There may, however, be some harmful effects also. If it is the most productive agricultural workers that leave, agricultural productivity may fall. There is also no guarantee that enough jobs will be available in urban areas and a number of external costs, including overcrowding, increased pressure on social capital and congestion may occur as a result of the rapid expansion of cities.

The effects of net emigration

There are a number of possible economic effects of net emigration. These are influenced by the size and the nature of the emigration. Some of these include:

- The size of the working population is likely to be reduced. Most emigrants tend to be of working age.
- The remaining labour force will have a greater burden of dependency.
- The average age of the labour force will increase. This may make it less mobile.
- The gender distribution of the population may be affected. In the case of some countries, more men emigrate than women.
- There may be a shortage of skilled workers. For example, doctors may emigrate in search of higher pay and better working conditions. The country they leave will experience a 'brain drain', while the country they go to will experience a 'brain gain'.
- There may be under-utilisation of resources. The country may become under-populated.
- Those who emigrate may send money home to help their relatives. This money is called workers' remittances.